Post-Classification Refinement and Statistical Analysis Workflow

1. Introduction: The Strategic Role of Post-Classification in Geospatial Analysis

In the field of remote sensing, the generation of a raw classified image is rarely the final stage of a project. Post-classification refinement is a strategic necessity that ensures data integrity and transforms raw computational outputs into actionable insights. Without these critical adjustments, classification maps often retain spectral artifacts and "speckle" noise that can obscure true spatial patterns and lead to inaccurate environmental assessments. The primary objective of this report is to define a structured methodology for cleaning classified maps, extracting rigorous quantitative statistics, and documenting spatial findings for professional use. By following this sequence, analysts ensure that their findings move beyond simple image processing to become high-quality geospatial intelligence. The process begins with essential refinement steps conducted immediately following the initial image classification to establish a baseline of accuracy.

2. Phase I: Image Refinement and Feature Identification

Before a classified map can be considered professional-grade, initial noise and spectral inconsistencies must be addressed. Raw output often contains isolated pixels or poorly defined features that do not accurately represent the target environment. Addressing these issues early in the workflow is essential for moving toward a clean, interpretable product. In this specific workflow, we treat these steps as residual noise isolation to ensure the spectral integrity of the classes.

The following three steps are critical for feature identification and initial refinement:

1. **Noise Removal via MNF (Minimum Noise Fraction) Rotation:** This process is utilized to isolate the signal from redundant or corrupt data. In a post-classification context, it serves to separate meaningful spectral information from the inherent noise of the capture, allowing for a cleaner identification of features.
2. **Finding Pure Points:** This step involves locating spectral endmembers—the pixels that represent the purest available sample of a specific material or land-cover class without sub-pixel mixing.

3. **Identify Ingredients (Mapping):** This process involves creating Regions of Interest (ROIs) for the image. Here, "ingredients" refers to the specific spectral classes or land-cover categories that define the map's legend.

Once these features are accurately identified and the spectral "ingredients" are mapped, the workflow shifts from feature identification to the spatial cleaning of the generated map to ensure pixel-level consistency.

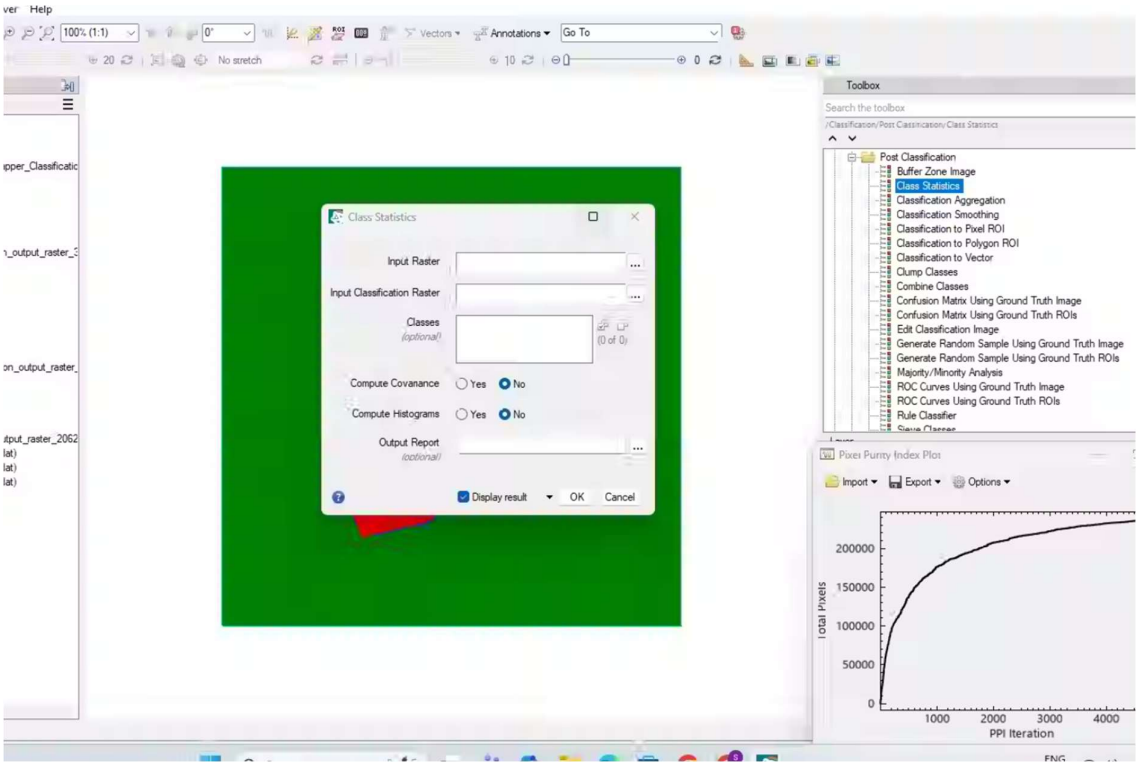
3. Phase II: Map Optimization through Majority Analysis (Step 4)

The "Step 4: Clean the Map" phase is designed to improve the spatial consistency of the classification. Raw maps often suffer from "speckle," where single, isolated pixels are classified differently than their surrounding neighbors. This spatial cleaning is a prerequisite for accurate statistics; without it, speckle noise can artificially inflate the pixel counts of minority classes, skewing the final data. By removing these isolated pixels, the analyst improves the overall interpretability of the final product and ensures that land-cover classes are represented as continuous, logical features.

To execute this optimization, navigate the software through the following path:

Toolbox > **Classification** > **Post Classification** > **Majority Analysis**

Upon selecting this tool, a parameter window will appear. Within this window, the analyst must define the search window or kernel size to determine the threshold for smoothing isolated pixels. This bridge between spatial cleaning and data extraction ensures the map is sufficiently refined for quantitative analysis. Following this optimization, the workflow proceeds to the extraction of class-specific data.



4. Phase III: Quantitative Data Extraction and Statistical Generation (Step 5)

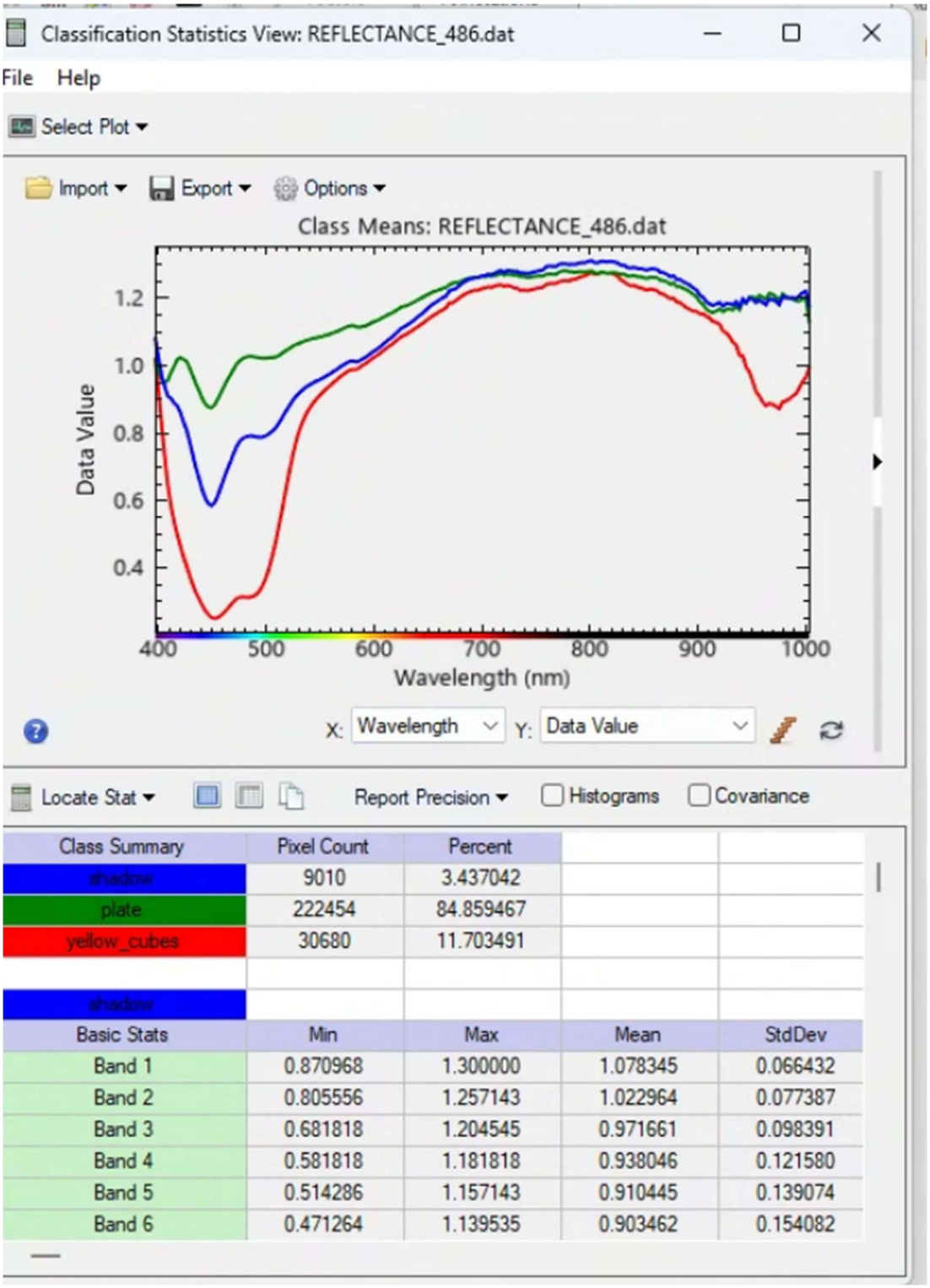
The strategic value of "Step 5: Extract the Data" lies in the transformation of raw pixel counts into meaningful percentages. This quantitative shift allows for rigorous comparative land-cover analysis, enabling decision-makers to understand the proportional distribution of features within a study area. Converting spatial data into statistical reports is the key step in turning an image into a document of record.

The data extraction process requires the following parameters to be defined:

- **Input Raster:** The original .dat image file.
- **Input Classification Raster:** The last step (**Step 3**) classified file. It is critical to use this specific file to ensure refinements are captured in the statistics.
- **Output File:** Assign a unique, descriptive name to the file for storage.

To access this function, follow the technical path: **Toolbox** > **Classification** > **Post Classification** > **Class Statistics**

After defining these inputs, the analyst must click "**OK**" to trigger the windows popup where the input classification map is confirmed for processing. The system then generates the specific findings required for the environmental report.



Class Summary	Pixel Count	Percent		
shadow	9010	3.437042		
plate	222454	84.859467		
yellow_cubes	30680	11.703491		
shadow				
Basic Stats	Min	Max	Mean	StdDev
Band 1	0.870968	1.300000	1.078345	0.066432
Band 2	0.805556	1.257143	1.022964	0.077387
Band 3	0.681818	1.204545	0.971661	0.098391
Band 4	0.581818	1.181818	0.938046	0.121580
Band 5	0.514286	1.157143	0.910445	0.139074
Band 6	0.471264	1.139535	0.903462	0.154082

5. Data Synthesis: Spatial Occupancy Results

The ultimate goal of this geospatial workflow is the precise quantification of the environment. In professional reporting, "Area Occupied" serves as the primary metric,

providing a clear breakdown of how much of the total study area is covered by each identified class.

Based on the statistical analysis performed during this process, the following spatial occupancy results were derived:

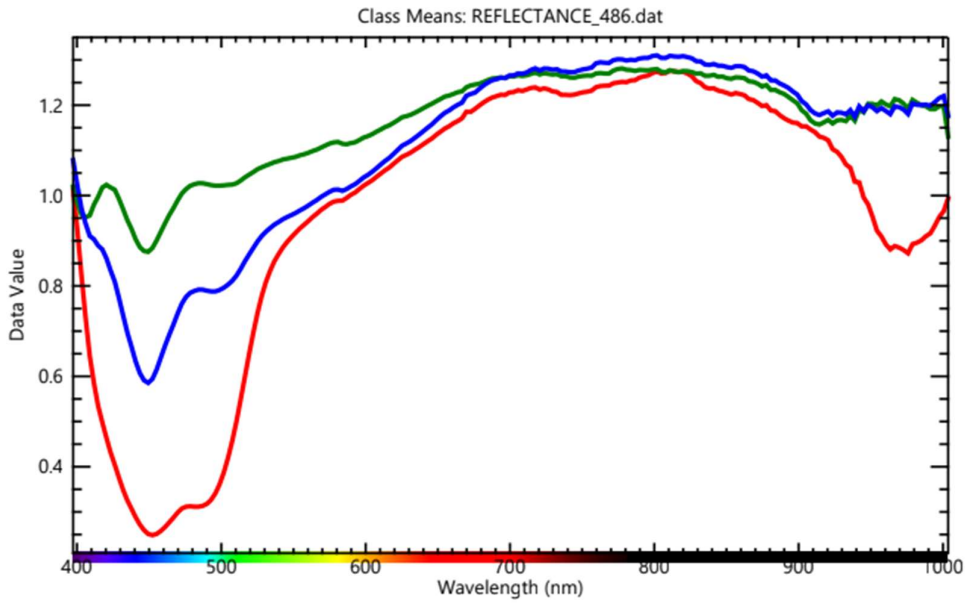
Class Name	Percentage Occupied (%)
Shadow	3.44%
Plate	84.9%
Yellow-Abs	11.7%

An analysis of these figures reveals that the "Plate" class is the dominant spatial feature within this dataset, occupying nearly 85% of the total area. The "Yellow-Abs" class represents a significant secondary feature at 11.7%, while "Shadow" accounts for a negligible fraction of the spatial footprint. With these figures established, the workflow moves to the final administrative steps of report preservation.

6. Phase IV: Documentation, Export, and Workflow Scalability (Step 6)

Standardized reporting is essential for maintaining a professional record of geospatial findings. Furthermore, preserving digital assets is critical for workflow scalability, allowing for efficient batch processing across multiple datasets in future projects.

To finalize the documentation and ensure the workflow is replicable, the following "Step 6" export instructions must be followed:

- **PDF Generation:** Within the report popup, navigate to **File** > **Save As** to finalize the documentation as a PDF file.
- **Batch Processing Preparation:** If you intend to perform batch processing on additional datasets, it is a critical requirement to **Save created ROI's as ros file**. This specific file format ensures that feature identifications and training data can be applied across multiple images with technical consistency.

By adhering to this structured workflow, analysts ensure both the technical accuracy and the professional utility of their geospatial reports, providing a clear, reproducible path from raw data to actionable spatial intelligence.